

Review

Diagnostic accuracy meta-analysis: A review of the basic principles of interpretation and application

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ABSTRACT

Evidence-based practice has become increasingly important in all aspects of medicine. The availability and variety of diagnostic tests in cardiovascular practice are increasing rapidly and are therefore under scrutiny in the literature. Diagnostic meta-analysis is used to summarise and combine results from multiple studies. The analytical techniques for diagnostic meta-analysis differ from other meta-analyses because the measures of diagnostic accuracy (sensitivity and specificity) are unique to diagnostics. However, the benchmark for diagnostic meta-analysis techniques continues to move forward, making interpretation and application of the results difficult for clinicians. This article explains the terminology of the current standards in diagnostic meta-analysis, and outlines the principles of applying these results to clinical practice.

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1. Introduction

Cardiovascular clinicians are regularly obliged to choose the most appropriate diagnostic tests for their patients, and as such diagnostic tests are a critical part of cardiovascular practice and patient outcomes. Evidence-based use of diagnostic tests is becoming more important and there is an increasingly large number of diagnostic tests available for cardiovascular practice. Clinicians must decide which test is most appropriate, depending on the accuracy, availability, ease of interpretation, healthcare costs and applicability to individual practice.

The primary focus of test accuracy is the evaluation of sensitivity and specificity. Sensitivity is the ability to correctly identify patients with the target condition, and does not provide information on the test's performance in patients without the condition. Conversely, specificity is the test's ability to correctly identify patients without the target condition, but does not give insight into its performance in patients with the condition. The sensitivity and specificity for each test can be used in comparison to those for other tests when deciding on which test to apply in a clinical setting.

Diagnostic test accuracy studies are being increasingly published in the medical literature [1]. Single well-designed studies can provide plausible and accurate results, however choices in study design, patient selection, disease spectrum and analytical methodology can limit the generalisability of the results to other patient populations or institutions. Multiple studies are therefore needed for a new diagnostic test to become widely accepted against a traditional test. Diagnostic meta-analysis can be used to obtain more precise estimates of test accuracy, by combining results from smaller studies evaluating the same diagnostic test in similar clinical settings [2]. Test accuracy is compared to that of a known reference test (or gold standard). These results are used to draw conclusions about test performance based on more data than is available in a single study. Meta-analyses may also identify patient or test factors which influence the test accuracy. This feature of meta-analysis allows the clinician to more appropriately choose a test in their individual practice setting. It is therefore increasingly recognised that diagnostic test accuracy meta-analysis is strong evidence in the assessment of diagnostic accuracy.

Over the past decade, there have been many developments in the methodology of diagnostic test accuracy meta-analysis [2]. These include more robust approaches to question formulation, literature searching, primary study quality assessment and the statistical analysis of the data. The terminology is rapidly changing and becoming more complex, making interpretation by clinicians more challenging. A basic understanding of diagnostic terminology and the principles of interpreting a diagnostic test accuracy meta-analysis is vital for optimal application to clinical practice. This article provides an overview of the principles of

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interpreting diagnostic test accuracy meta-analysis results, and will help readers correctly appraise and apply the evidence to their own practice. Diagnostic accuracy meta-analyses from the cardiovascular literature will be used to illustrate these principles. The specific practicalities of performing a meta-analysis are well described elsewhere [3–5], and the aim of this manuscript is to equip the reader with the concepts required to apply these meta-analytical results in cardiovascular clinical practice.

2. Underlying principles of diagnostic accuracy meta-analysis – clinical question and literature search

Diagnostic meta-analysis is performed via a series of steps which remain consistent (Table 1). Careful adherence to the meta-analytical process prevents sloppy methodology and increases the reliability of the analysis conclusions. Table 2 gives the explanations of commonly used terms in diagnostic meta-analysis.

The first step in performing a diagnostic accuracy meta-analysis is the formulation of a clinical question. At this stage, there should be clear definitions of the diagnostic test under evaluation (the index test), the reference test (also known as the gold standard) against which the index test is compared, and any alternative tests which are included for comparison. The clinical question also specifies any relevant population and disease characteristics. A decision about what constitutes appropriate studies will depend on relevant population characteristics (including disease prevalence, spectrum of disease and inclusion and exclusion patient criteria), the availability of prior test information, the diagnostic threshold(s) used in the studies, and any other factors which are known from the literature to influence the test accuracy. In addition, there must be sufficient data in the primary study for derivation of the accuracy parameters required for the analysis.

When defining the index test, it is important to specify the test characteristics, including equipment factors, scanning protocols and methods of interpretation. An example of this is found in a recently published meta-analysis of multidetector computed tomography (MDCT) in the evaluation of acute chest pain [6]. In this example, the index test is MDCT, with information extracted on number of CT detector rows and scanning protocols, including the administration of beta blockers. Clearly, these factors may influence the accuracy of the test and should be documented.

The reference test should also be defined at this stage, with detailed reasons for patients undergoing different reference stan-

Table 2

Summary of terms used in diagnostic meta-analysis.

Term	Explanation
Heterogeneity	Variation in results between studies
Covariate	A characteristic of the study, patients, test or something else, which could be influencing the test accuracy
Diagnostic odds ratio (DOR)	A summary estimate of diagnostic accuracy, incorporating both sensitivity and specificity. Used primarily in meta-analysis to provide a statistic to allow comparison between different tests. It is calculated by $(TP \times TN) / (FP \times FN)$.
Area under the curve (AUC)	The area under the SROC curve. A unitless number between zero and one inclusive. The closer the value to one, the greater the accuracy of the test. The value of AUC depends on the method of curve generation (SROC or HSROC).
Bias	Discrepancy between the true and observed values of the outcomes of interest, due to aspects of study design, patient selection, observer technique or analysis methods.
QUADAS	A 14 point tool which is used to highlight areas of bias within the primary studies included in a diagnostic meta-analysis.
Diagnostic threshold	The point at which a result is considered positive.

dards. Ideally all studies in the meta-analysis use the same reference test, and within each study all patients undergo the same reference standard. However depending on the literature available, this may not be feasible. In the above example [6], the inclusion criteria for the meta-analysis allowed reference standards including coronary angiography, present standard of care and clinical follow up. This leads to different definitions of a positive outcome. In patients undergoing coronary angiography, an occlusion or stenosis of at least 50% of a coronary artery was considered a positive result. In other patients, a positive result was clinical unstable angina pectoris, non-ST elevated myocardial infarction, or major adverse cardiovascular events (death, acute myocardial infarction or unstable angina) during the index visit or follow up period. Whilst not ideal, this situation is common in the literature as many diagnostic accuracy studies are retrospective, and the ideal reference standard (in this example, coronary angiography) is not without morbidity and risk. Meta-analyses which include studies with widely varied reference standards introduce additional sources of heterogeneity (variation in results) and attempts can be made to account for this in the analysis.

In some circumstances, the reference standard is in routine use and the index test is the only proposed alternative test. However as there are increasing numbers of diagnostic tests available, the more usual situation is for a number of other alternative tests to be available. Comparative meta-analyses compare the performance of the index test to alternative testing strategies, and therefore provide more information for both clinicians and policy-makers about the most appropriate choice of test. For example, a recent meta-analysis directly comparing the accuracy of dipyridamole and dobutamine stress echocardiography, against the same reference standard of coronary angiography, was performed with the explicit intention of determining whether either of the protocols performed substantially better than the other [7]. This comparative approach gives more information for the clinician than a meta-analysis assessing the accuracy of one protocol only.

Disease prevalence in the included studies also influences test accuracy. In the cardiac MDCT example [6], the authors examine the usefulness of MDCT in patients with unclear aetiology of chest pain in whom management strategies are unclear. They include patients with low to intermediate risk of acute coronary syndrome (ACS), and exclude patients with definite myocardial infarction or high risk of ACS. This means that the prevalence of ACS in the included studies is less than in all patients presenting with acute chest pain. The test accuracy is therefore only applicable to patients with this presentation. The disease severity (or spectrum of disease) is also important. Patients with severe disease are easier to identify than those with mild to moderate disease, potentially overestimating the test accuracy.

Table 1

Summary box of the steps of diagnostic test accuracy meta-analysis.

Step	Details
Define the clinical question	Define the index test, reference test, any comparison test, disease spectrum and prevalence, patient factors, diagnostic threshold, prior test information, data availability
Literature search	Written record of search terms, databases searched, dates of search. Use of synonymous terms if necessary. Manual reference check if appropriate.
Assessment of methodological quality	Use of a checklist such as QUADAS to identify potential sources of bias and account for its effect on the end results. Additional biases considered where appropriate.
Data analysis	Robust analytical methods, with results presented in a manner understood by clinicians. If multiple tests are being compared to a reference test, direct comparison between the tests may be better facilitated by graphical results.
Applicability	Drawing conclusions about the applicability of the results – discussion of the limitations and strengths of the meta-analysis, caveats for interpretation and identification of subgroups where the results must be interpreted with caution.

Disease severity is also dependent on the setting of the study (for example secondary or tertiary care) and this may influence the results and should be taken into account in the interpretation of the test results.

In addition to disease prevalence and severity, other patient characteristics may influence the test accuracy. For example, the presence of cardiac arrhythmia may alter the accuracy of MDCT in coronary artery bypass graft assessment, and meta-analysis of this topic includes data extraction on the inclusion of patients with arrhythmias [8]. This sort of information can be included in the analysis as a covariate (factor which may influence the accuracy) (Table 2).

The choice of diagnostic threshold (the point at which a test result becomes positive) will determine the sensitivity and specificity of the test. In many examples in the literature there are varying definitions of a positive test result. In the MDCT example, significant coronary artery stenosis was defined as greater than 50% luminal diameter narrowing or complete occlusion [6]. Other studies in the literature may define significant stenosis as 70% or 75%. Even when the reported thresholds are constant between studies, the true thresholds will differ due to individual variance in test interpretation. For this reason, it is possible to include studies with different thresholds in the same meta-analysis, provided there is a clear rationale for these differing thresholds.

For most diagnostic meta-analyses, the statistical endpoints are sensitivity and specificity. For this reason, it must be possible to extract or calculate values for both these statistics from the included studies. Individual patient data are not currently required for most diagnostic meta-analyses. However, in cases where there are a small number of studies or a large number of covariates, individual patient data are used to increase the robustness of the methodology.

After careful formulation of the clinical question, a thorough search of the literature is performed for the primary studies. This searching is hindered by the lack of common keywords for diagnostic accuracy studies, although the Medical Subject Heading 'sensitivity and specificity' is sometimes used [2]. An additional problem is that diagnostic accuracy results may be reported in studies which do not refer to that part of its content in the abstract, making those results difficult to identify. Obviously, if there are multiple tests being compared, or a single test with a number of duplicate names, the number of search terms increases. For example, in cardiovascular literature the terms 'coronary', 'cardiac' and 'heart' should all be used in the search strategy. Similarly, 'multidetector' and 'multislice' CT are synonymous terms. A written record should be kept of the search strategy, with dates of search and the exact search terms recorded for each database. An example of this search technique is a Heath Technology Assessment review of cardiac CT in the assessment of coronary artery disease, where a comprehensive search strategy is documented in an appendix [9]. This should be exhaustive, and not limited by publication language. A manual reference check, as well as searching in conference listings, can be performed to ensure all relevant articles are identified.

3. Quality assessment

Variation in results between studies evaluating the diagnostic accuracy of the same test against the same reference standard across the literature is expected. Identifying reasons for heterogeneity is partly why meta-analysis is so useful in evaluating overall test accuracy. Reasons for this variation (or heterogeneity) are differences in study design, setting of the study and study population, disease spectrum, disease prevalence, test characteristics, diagnostic threshold, and availability of other test information, as well as random variation. Random variation is more likely when the primary studies have small sample sizes [10].

The quality of the meta-analysis partially depends on the methodological quality of the primary studies [11]. If a primary

study has a flawed design, the results may be biased. Bias refers to differences between the observed and true (unknown) values of the study endpoints. It is not measured directly, but robust methodology and endeavors to minimise known sources of bias increase the credibility of observed results. The type of bias and its effect on the validity of the results depend on the nature of the design flaw. A summary of the more common and important types of bias is provided below and in Table 3. Generally, there are two broad categories of bias, measurement or information bias and selection bias. Assessing any diagnostic accuracy study for bias, not just in the context of a meta-analysis but in general, will improve the reader's ability to appropriately apply the results to their own practice.

3.1. Measurement bias

The reference test should correctly classify the target condition. When this is not the case, misclassification bias (or reference standard error bias) occurs. An example of this is coronary angiography in the assessment of coronary artery bypass graft stenosis, where inability to visualize the graft lumen is interpreted as occlusion, but may be due to technical factors [8]. The entire study population, or a randomly selected subgroup, should undergo the same reference test. Partial verification bias (or workup bias) occurs when a non-random selection of the study population undergoes the reference test. This is common in the assessment of acute chest pain, where a negative stress test may prevent a coronary angiogram being performed.

Differential verification bias occurs when the reference standard changes with the index test result. This is the case in some primary studies in the acute chest pain meta-analysis above [6], where the reference standard depended on the MDCT result. Incorporation bias occurs when the index test result contributes to the final reference test result, making the concordance rate between the two tests artificially high.

The time between the diagnostic and reference tests should be sufficiently short to prevent significant disease progression between results (disease progression bias). When one test is interpreted with awareness of the other test result (review bias), accuracy tends to be overestimated. For example, if carotid angiography is performed with knowledge of the results of carotid artery ultrasound or CT angiography, it is more likely that the results will agree. This may be acceptable in clinical practice, but is undesirable in impartial assessment of the accuracy of a diagnostic test.

Clinical review bias occurs when there is a subjective or interpretive component to the test, allowing clinical information to contribute to the final diagnosis. If this occurs, the methodology should explicitly state this to enable accurate adoption of the technique elsewhere. If equivocal or unavailable results are excluded, bias can occur if the remaining study population differs significantly from the initial group.

3.2. Selection bias

Spectrum bias occurs when the included patients do not accurately reflect the intended disease spectrum for the target population. For example, cardiac magnetic resonance imaging is more likely to correctly diagnose left ventricular failure in patients in New York Heart Association categories III and IV than those in categories I and II. Any non-random patient selection should be stated in the inclusion criteria so that the results are not inappropriately applied to the general population.

The inclusion of "normal" patients as controls, and known severe cases as the 'positive' cases overestimates test accuracy as the extremes of health are generally easier to detect. Similarly, excluding patients with diseases known to adversely affect the diagnostic performance of the test leads to limited challenge bias, as in the

Table 3
Sources of bias commonly encountered in studies of diagnostic test accuracy.

Bias type		Methodological flaw leading to bias
Measurement bias		
Reference test	Misclassification	Incorrect result on the reference test.
	Partial verification	A non-random selection of patients undergoes the reference test.
Index or reference test	Differential verification	The reference test depends on the result of the index test.
	Incorporation	The index test result contributes to the reference test result.
Selection Bias	Review	One test is interpreted with prior knowledge of the other test result.
	Progression	The disease severity progresses in the time between the index and reference tests.
	Clinical review	Clinical information contributes to the test result, particularly if the test is subjective.
	Disease spectrum	The included patients do not reflect the target population in terms of disease spectrum.
	Patient selection	Non consecutive or non-random inclusion of eligible patients.

example of patients with chronic obstructive pulmonary disease excluded from a study of ventilation/perfusion scintillation scanning.

Assessment of the primary study quality as part of a diagnostic accuracy meta-analysis is therefore performed to identify sources of bias, which can be accounted for in the analysis and their effect on the results minimised [2]. QUADAS (Quality Assessment of Diagnostic Accuracy Studies) [12] is a checklist of 14 items often used in diagnostic accuracy meta-analysis to assess the methodology of the included studies. If additional sources of bias apply to a meta-analysis, it may be necessary to add extra sources of biases into the quality assessment. Similarly, one or more QUADAS items may be irrelevant and be omitted. Each analysis should incorporate the relevant biases, and QUADAS should be considered a starting point rather than a rigid checklist. Rather than using an overall bias score, each bias (assessed by an item in the tool) should be examined separately, otherwise important sources of bias can be masked. Most biases in diagnostic accuracy meta-analysis tend to overestimate test accuracy, particularly the inclusion of healthy controls and the use of different reference standards in non-random selections of the study population [2].

Different methods of quantifying the effect of each source of bias have been detailed elsewhere [3,13]. Separate univariate regression analysis of each item, with subsequent multivariate modeling of influential items (with $p < 0.10$ in the univariate model) is one method advocated to assess their influence on the results [11]. The influence of the variable (or covariate) is expressed as the relative diagnostic odds ratio (RDOR). RDOR is calculated as the DOR when the covariate is positive, as a ratio to the DOR when the covariate is negative. Westwood et al [11] provide a thorough discussion and worked examples of univariate and selective multivariate analysis of QUADAS tool items, as well as carefully chosen analysis-specific covariates. An example of this method is meta-analysis of coronary bypass graft assessment with multislice CT, where availability of clinical data at time of reporting was associated with increased accuracy in diagnosis of graft stenosis on CT [8].

Sensitivity analysis is another method available for evaluating the effect of bias in poor quality studies on the overall results of the meta-analysis. This is achieved by comparing the results obtained when including versus excluding the study (or studies) incorporating that type of bias. This allows inclusion of the studies into the meta-analysis, whilst still acknowledging their flaws. This is however difficult to implement when the number of studies eligible for inclusion into the meta-analysis is small.

4. Diagnostic accuracy estimation

The reference test result is generally assumed to indicate the true disease status. The index test gives a true positive (TP), true negative (TN), false positive (FP) or false negative (FN) in comparison with the reference standard result. A two by two table is often used to help estimate these numbers (Fig. 1a). For example, in the MDCT meta-

analysis [6], the primary study by Hoffmann et al. [14] included 103 patients, of which 14 had acute coronary syndrome (ACS). All 14 were correctly identified on MDCT, giving sensitivity of 14/14 (100%). However, 16 other patients in whom significant stenosis could not be excluded did not have ACS, creating 16 false positive results. Of 89 patients without ACS on reference standard testing, only 73 were negative on MDCT (73 true negatives), giving specificity 73/89 (82%) (Fig. 1b).

Sensitivity and specificity are measures of diagnostic test accuracy. Sensitivity is the ability to correctly identify diseased patients, but does not give information about the test's performance in healthy patients. It is the ratio of true positives (TP) to all positive reference standard test results (TP + FN) (Fig. 1a). Specificity is the ability to identify healthy patients, but gives no information on diseased patients. It is the ratio of true negatives (TN) to all negative reference standard results (FP + TN) (Fig. 1a). In diagnostic meta-analysis, sensitivity is known as true positive rate (TPR), and (1-specificity) is known as false positive rate (FPR).

Sensitivity and specificity are given as a pair of values to help clinicians choose the most appropriate test, but depend on study design, diagnostic threshold and disease prevalence in the study population.

The diagnostic odds ratio (DOR) is a summary estimate of test accuracy commonly used in diagnostic meta-analysis. It is the odds of positive test results in patients with disease, compared with the odds of positive test results in those without disease, calculated as $(TP/FN)/(FP/TN) = (TP \times TN)/(FN \times FP)$. A higher value of DOR indicates greater overall

a				
		REFERENCE STANDARD		
		+	-	TOTAL
INDEX TEST	+	TP	FP	TP + FP
	-	FN	TN	TN + FN
	TOTAL	TP + FN	TN + FP	TP + FN + FP + TN
b				
		REFERENCE STANDARD		
		+	-	TOTAL
MDCT	+	14	16	30
	-	0	73	73
	TOTAL	14	89	103

Fig. 1. a. The two by two table commonly used to calculate true positives, true negatives, false positives and false negatives. Sensitivity, specificity and predictive values can be easily calculated using this table. b. The example data from Hoffmann et al. [14] in a two by two table.

accuracy. It is used to compare the overall accuracy of different tests, but is rarely used in clinical practice as it is not as clinically meaningful as sensitivity and specificity. There may be a need for a test with either high sensitivity or high specificity, but not necessarily both. Two tests may have the same DOR, but have differing sensitivity and specificity, and thus one may be preferred over the other on clinical grounds. DOR does not give information on the different types of false results.

Positive and negative predictive values (PPV and NPV) give the probability of a known test result being correct. The PPV and NPV are highly dependent on disease prevalence in the study population, and are therefore less useful than sensitivity and specificity in evaluating inherent test accuracy. PPV is the ratio of true positives (TP) to all the positive index test results (TP + FP). NPV is the ratio of true negatives (TN) to all negative index test results (TN + FN). PPV and NPV are not usually analysed in diagnostic meta-analysis, but their differences from sensitivity and specificity should still be clearly understood.

5. Combining results from multiple studies

The simplest method of summarising multiple results is averaging them out (pooling). However, this does not account for differences in study size, thresholds, or patient characteristics. Wildly varying data points can average out with misleading results. It also does not take into account the relationship that exists between sensitivity and specificity. For these reasons, pooling is not recommended for diagnostic accuracy meta-analysis.

Summary receiver operating characteristic (SROC) analysis was suggested by Moses et al [15]. SROC is an extension of receiver operative characteristic (ROC) analysis which has been previously described [16]. The modelling behind SROC results uses both logarithmic transformations and linear regression models and more detailed information is available elsewhere [15,17]. SROC produces a summary DOR value to estimate the test accuracy against the reference standard. As DOR incorporates both sensitivity and specificity into the one value, the relationship between them is also accounted for in the SROC model. However, it is the SROC graph which is often confusing for clinicians. The stress echocardiography meta-analysis briefly described above will be used to demonstrate the graphical methods of showing diagnostic accuracy meta-analysis results. Picano et al. [7] examine the comparative accuracy of dobutamine versus dipyridole stress echocardiography in the diagnosis of coronary artery stenosis, using coronary angiography as the reference standard.

A forest plot can be used to graphically show the spread of sensitivity or specificity values for the studies included in the meta-analysis. The sensitivity and specificity data from the five studies included in the stress echo meta-analysis [18–22] have been shown in Fig. 2 in a forest plot.

Each (sensitivity, 1-specificity) pair from the primary studies is then plotted onto a graph termed the SROC scatterplot. The graph has a horizontal axis measuring FPR (1-specificity) and a vertical axis measuring TPR (sensitivity). Both axes span the range of (0, 1). The data for the dobutamine stress echo accuracy are plotted over the SROC space in Fig. 3a. This shows the variation in results, with the preponderance of data points in the upper left hand corner of the plot subjectively indicating good diagnostic accuracy.

A curve is graphed over the scatterplot, the equation of which is derived from linear regression of variables based on logit transformations of TPR and FPR. Fig. 3b shows the SROC curve superimposed over the scatterplot for the dobutamine data. In addition to estimating a summary value for DOR, the area under the SROC curve (AUC) is calculated. AUC measures the closeness of the index test to the reference test, which is assumed to be flawless. An ideal diagnostic test has a curve following the perfect square, giving TPR of 100% for every value of FPR. The closer the AUC value to one, the better the accuracy. The AUC for the dobutamine data is 0.93, indicating very good diagnostic accuracy.

The data in the stress echo example can also be used to compare the two different test protocols. Fig. 4 shows the two SROC curves for the dobutamine and dipyridole protocols drawn on the same set of axes. The curves are very similar, indicating that there the two tests are similar in diagnostic accuracy.

Whilst weights such as study size and variance have been used, they give disproportionate emphasis to smaller studies. These smaller size studies are more likely to give inaccurate results [5,15]. The calculations for SROC are performed by software packages which allow the user to select weights if desired.

There are many limitations of the Moses approach. A summary estimate of either sensitivity or specificity can be produced, but only after one of them has been allocated a value by the meta-analyst. Confidence intervals for the sensitivity and specificity estimates cannot be produced, further limiting their usefulness. DOR and its confidence interval can be used to summarise overall accuracy, but the clinical scenario may favour a test with high sensitivity or specificity. The Moses model does not account for different numbers of diseased and non-diseased patients between the included studies [17]. Weighting by overall study size does not give accurate weighting to

Dipyridole Accuracy vs CT

Study	TP	FP	FN	TN	Sensitivity	Specificity
Loimaala	41	4	3	12	0.93 [0.81, 0.99]	0.75 [0.48, 0.93]
Nedeljkovic	66	4	3	44	0.96 [0.88, 0.99]	0.92 [0.80, 0.98]
Pingitore	75	1	17	17	0.82 [0.72, 0.89]	0.94 [0.73, 1.00]
Salustri	23	2	5	16	0.82 [0.63, 0.94]	0.89 [0.65, 0.99]
San Roman	54	2	12	34	0.82 [0.70, 0.90]	0.94 [0.81, 0.99]

Dobutamine Accuracy vs CT

Study	TP	FP	FN	TN	Sensitivity	Specificity
Loimaala	42	6	2	10	0.95 [0.85, 0.99]	0.63 [0.35, 0.85]
Nedeljkovic	64	4	5	44	0.93 [0.84, 0.98]	0.92 [0.80, 0.98]
Pingitore	77	2	15	16	0.84 [0.75, 0.91]	0.89 [0.65, 0.99]
Salustri	22	4	6	14	0.79 [0.59, 0.92]	0.78 [0.52, 0.94]
San Roman	52	4	14	32	0.79 [0.67, 0.88]	0.89 [0.74, 0.97]

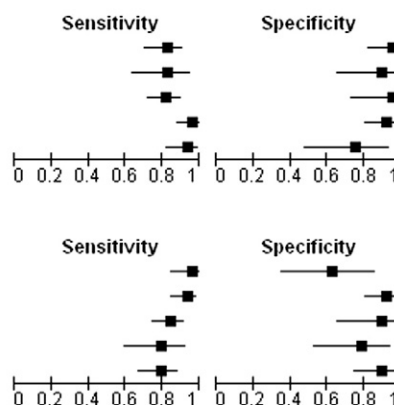


Fig. 2. Forest plot of sensitivity and specificity for the primary studies included in a comparative meta-analysis of dipyridole and dobutamine stress echocardiography accuracy compared to coronary angiography in the assessment of coronary artery stenosis [7].

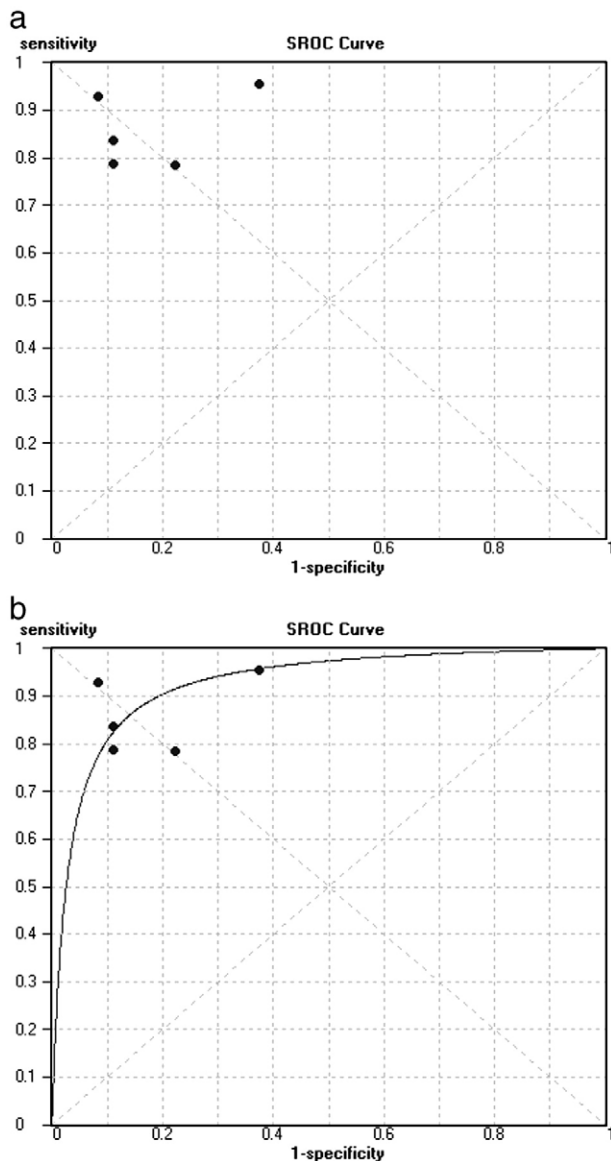


Fig. 3. a. SROC scatterplot of the dobutamine stress echo data [7]. Each point represents a (TPR, FPR) data pair from a primary study. b. The scatterplot in 3a with the SROC curve superimposed.

the TPR and FPR estimates, as these depend on the number of positive and negative results. The linear regression approach is too simplistic. The covariates that can be entered into the model are limited to factors which vary between studies, meaning that intra-study factors cannot be included.

6. Hierarchical SROC

The hierarchical approach to SROC (HSROC) addresses the limitations of the Moses model. There are currently two levels of modelling, using a binomial distribution in the first (intra-study) level, and a normal distribution in the second (inter-study) level [23]. This model produces a similar curve to SROC curve, but summary estimates of sensitivity, specificity and DOR along with their confidence intervals can be produced. However, the HSROC model is more complex to perform [23,24].

From the clinician's perspective, HSROC data are expressed with similar methods to SROC. The main difference is that the results are more robust, allowing greater investigation of factors which affect test

accuracy. It also allows direct comparison of different tests in the same studies, so that the study characteristics are constant for both tests. HSROC is more flexible than SROC in explaining heterogeneity, with both intra- and inter-study covariates allowed in the model.

7. Applying meta-analysis results to clinical practice

Reading and interpreting the results of a diagnostic meta-analysis can be difficult for the clinician. Correctly applying those results to one's own practice is likely to be more challenging still. There are many aspects of the analysis to consider, but the main ones are briefly discussed here.

The population of patients included in the primary studies should be similar to at least part of the clinician's own practice. The inclusion and exclusion criteria of the meta-analysis should be examined for evidence of particular population characteristics. For example, if a meta-analysis (or its primary studies) of coronary artery MDCT excluded patients with arrhythmias, the results do not directly apply to real life patients with arrhythmias. Similarly, if a meta-analysis excludes patients by age, gender or previous investigation result, these results are not applicable to those patients in one's clinical practice.

The spectrum of disease should also be considered. If the primary studies evaluate the extremes of the spectrum, the test accuracy will likely be overestimated and the results will not be applicable to the spectrum of disease in clinical practice. It is difficult to determine whether the disease spectrum of the primary studies is similar to one's practice without reading them, but examining the inclusion and exclusion criteria of the meta-analysis and any tables with patient characteristics may be helpful.

It is equally important to consider whether the test being examined is available in the practice setting. This may require liaison with other specialties (such as radiology or pathology) or discussion about initiating a new service. If the test is available, assess whether the test is performed locally in a similar way to the primary studies. For example, magnetic resonance imaging is performed at different centres with different field strength, coils, and sequence protocols. Computed tomography units possess varying numbers of detectors and temporal resolution.

It is important to consider whether the diagnostic pathway described in the meta-analysis can be directly compared to one's own practice. For

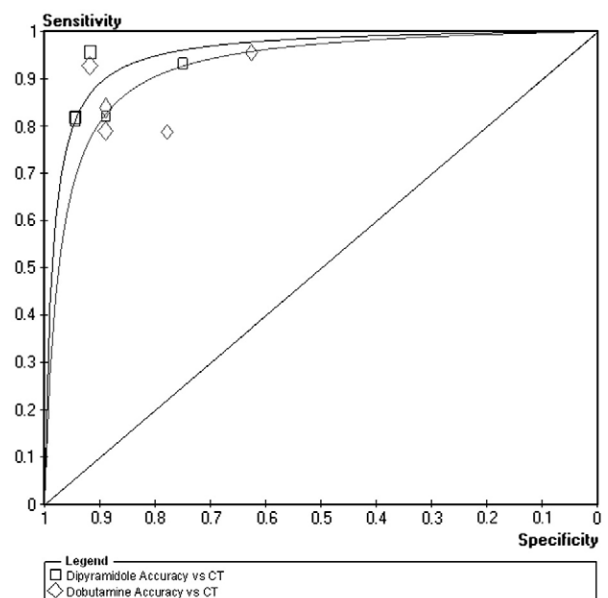


Fig. 4. The two SROC plots for the dobutamine (diamond) and dipyrindole (square) data. The sizing of the data point shapes reflects the original study size.

the MDCT acute chest pain example [6], only cardiovascular clinicians managing patients in the emergency room using the same diagnostic pathways as the primary studies will find the results directly applicable to their practice. Similarly, if the meta-analysis is comparative, but the local diagnostic pathway consists of an alternative other than that used in the meta-analysis, then no direct comparison with the existing local pathway is possible. If the index test is being used as a triage or add-on test, rather than as a replacement, it is vital to consider whether the rest of the pathway in the meta-analysis is sufficiently similar to that in the local setting. Some meta-analyses report on cutting edge technology which is not yet available in the wider medical community. Other meta-analyses may include older technologies which are no longer in use, such as 16 slice CT or single head gamma cameras.

Finally, another aspect of applicability is evaluation of the quality of the analysis. This incorporates the quality assessment of the primary studies, which may involve exclusion of poor quality studies, use of the QUADAS tool items as covariates, or sensitivity analysis to assess the influence of poor quality studies on the overall results [1]. The inclusion of all potentially important contributing factors (covariates) to test accuracy should be sought. For example, if heart rate, previous stenting, bypass graft surgery, or the use of rate lowering agents was not commented upon in a meta-analysis of coronary artery CT, the reader would be left wondering why. The inclusion of all available literature should be clearly outlined in the analysis, with the results of each search stated. Exclusion of study results should be justified. Use of HSROC instead of SROC indicates a more robust underlying methodology.

8. Conclusion

Diagnostic test meta-analysis is the cornerstone of clinical decision making. The reliance on high quality diagnostic meta-analysis is therefore increasing. This high quality requirement means that quality assessment of the primary studies with a recognised tool such as QUADAS should be done to identify sources of results' heterogeneity of and highlight areas of concern in study reporting standards.

The SROC approach incorporates the relationship between sensitivity and specificity, and has been extensively used over the past decade. Results from meta-analyses using this method continue to be published. However, its failure to generate estimates of the most clinically relevant measures, sensitivity and specificity, with any confidence intervals means that it has relied upon less well known statistics like AUC and DOR. From the clinician's perspective, applying these results to clinical practice can be difficult. For this reason, as well as methodological reasons, the HSROC approach is superior. The advances in HSROC modelling and the increasing availability of software make diagnostic meta-analysis more useful for clinicians. The terminology and methods of diagnostic meta-analysis are confusing for those unacquainted with meta-analytical techniques, but once the basic principles are known, diagnostic meta-analysis results can be safely interpreted and applied to one's own practice.

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The authors of this manuscript have certified that they comply with the Principles of Ethical Publishing in the International Journal of Cardiology [25].

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